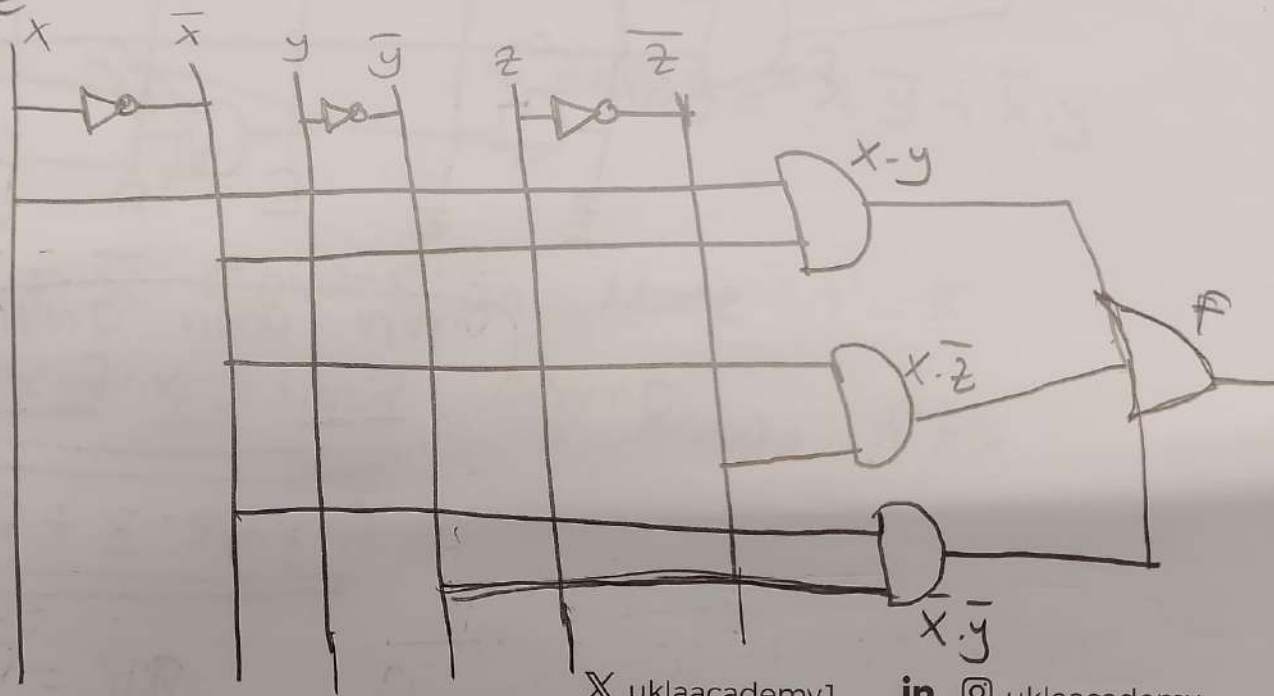


Q: $f = \underline{x \cdot y} + \bar{x} \cdot \bar{z} + \bar{x} \bar{y}$

FC (x, y, z)

A) Normalce



X uklaacademy1

in @ uklaacademy

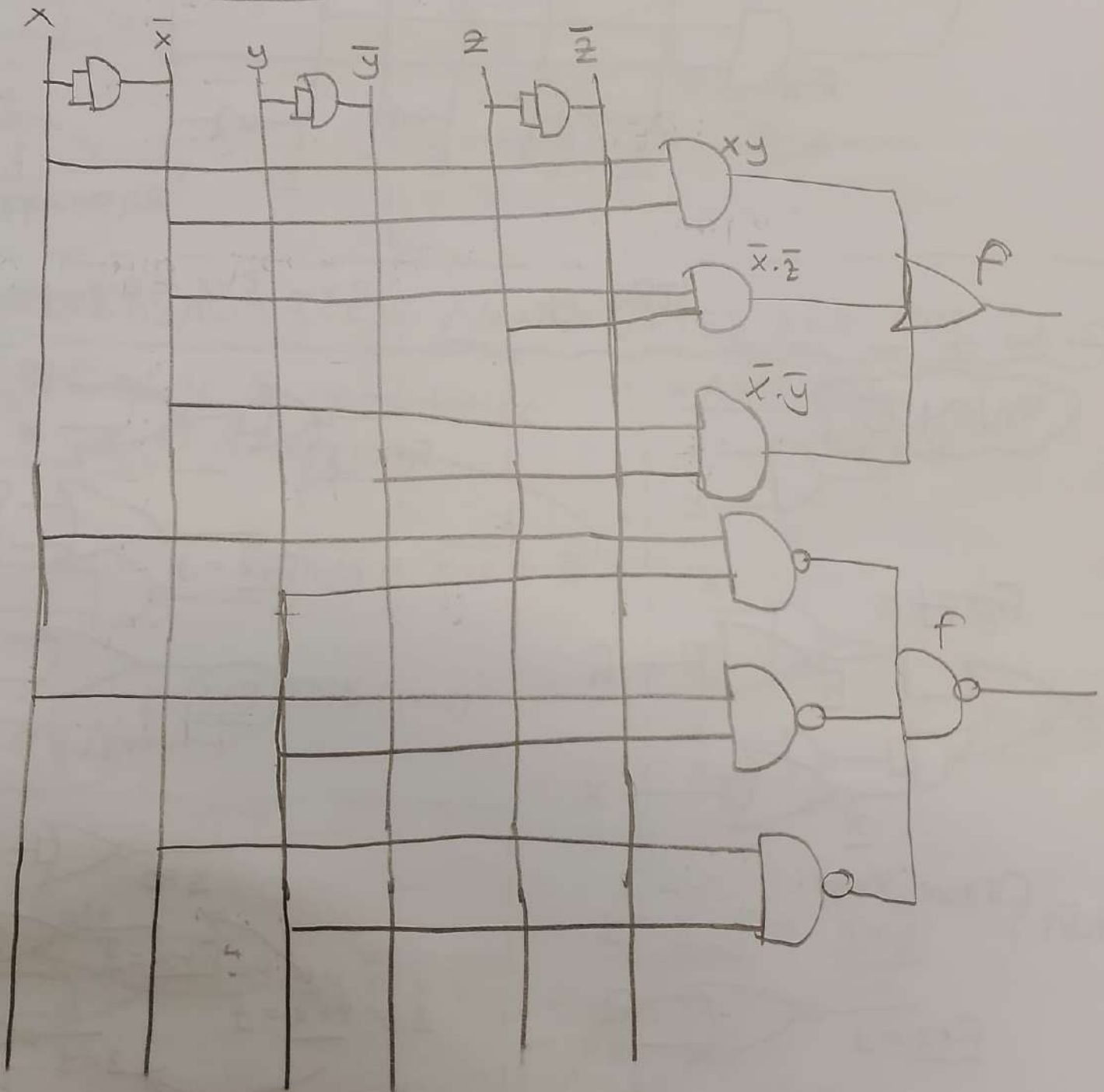
B) Yalnızca NAND kullanarak yap

$$f = \overline{x \cdot y} = \overline{AB}$$

$$\overline{f} = \overline{x \cdot y + \overline{x} \cdot z + \overline{x} \cdot y}$$

$$= \overline{x \cdot y} \cdot \overline{x \cdot z} \cdot \overline{x \cdot y}$$

NAND NAND NAND



OP: XOR

Dec	x y	f = x ⊕ y
0	0 0	0
1	0 1	1
2	1 0	1
3	1 1	0

$$f = x \cdot \bar{y} + \bar{x} \cdot y = x \oplus y$$



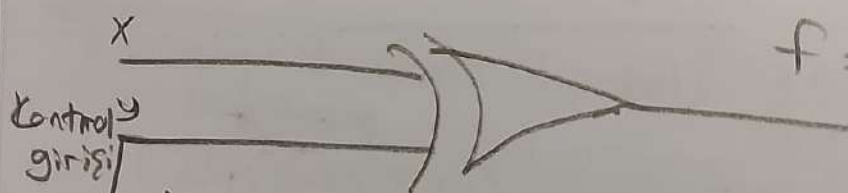
OP: XNOR

Dec	x y	f = x ⊙ y
0	0 0	1
1	0 1	0
2	1 0	0
3	1 1	1

$$f = x \cdot y + \bar{x} \cdot \bar{y} = x \odot y$$



Kontrollo Logic Seviye Dönüşümü



$$f = x \oplus y = x \cdot \bar{y} + \bar{x} \cdot y$$

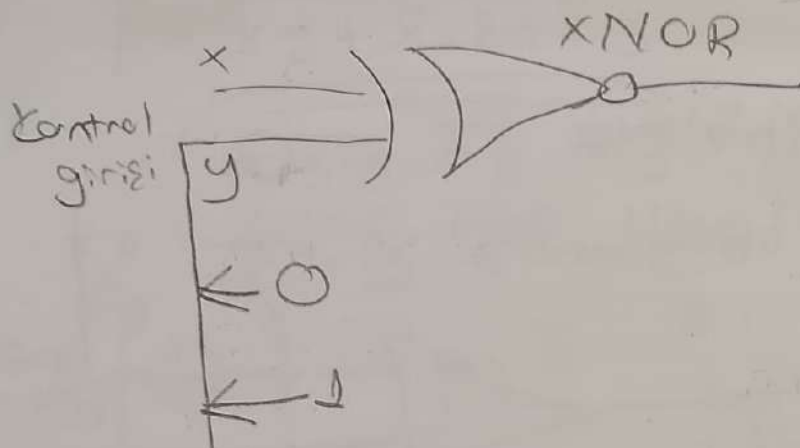
1 = $\bar{x}$ Conversion Mode $f = \bar{x}$

0 = x Transfer Mode $f = x$



0 = $\bar{x}$ conversion Mode $f = \bar{x}$

1 = x transfer Mode $f = x$

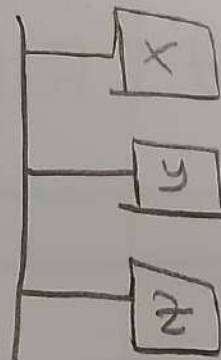
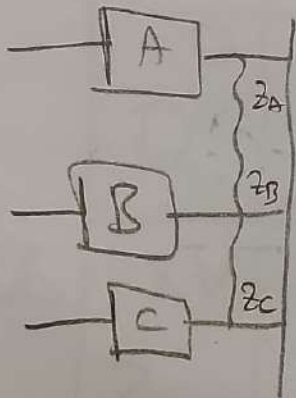


Buffer: Sinyallerin Seviyesini deđistirmeden ideal hale getiriliyor.

x	y
0	0
1	1



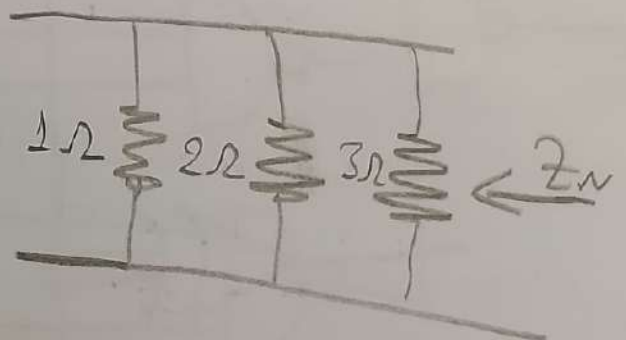
Tristate Gates

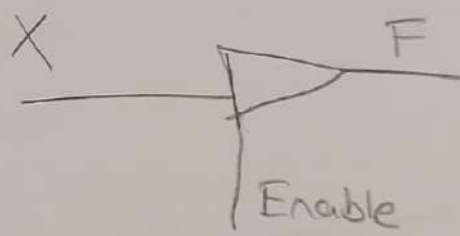
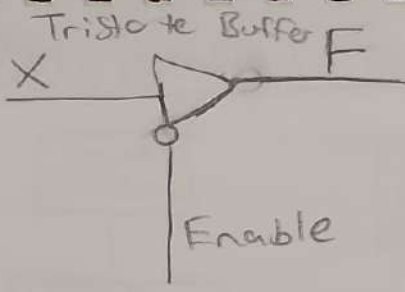


Common Bag

Common Bag

$$1 = \frac{1}{z_A} + \frac{1}{z_B} + \frac{1}{z_C}$$

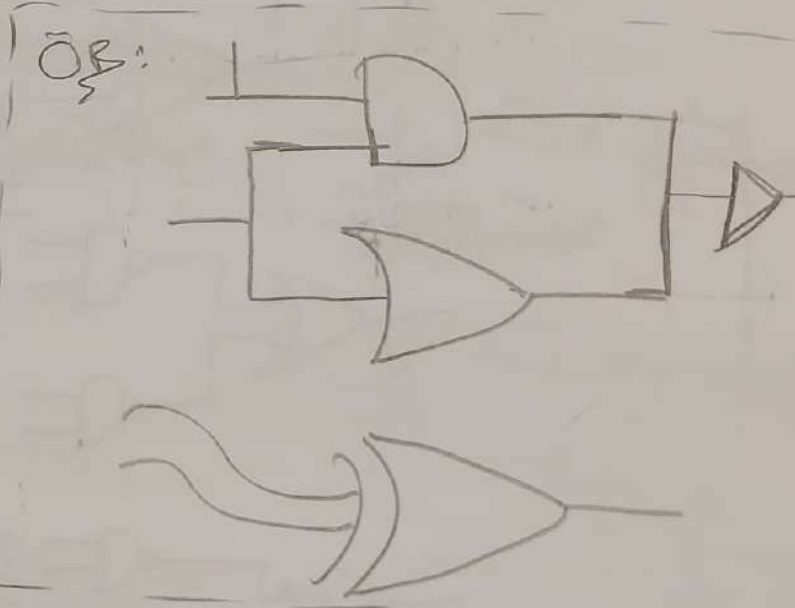
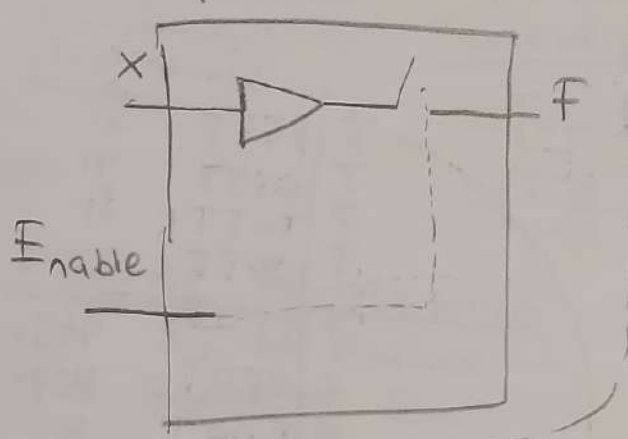




E	F
0	X
1	Hi-Z

↖ yüksek empedans

E	F
0	Hi-Z
Hi-Z	X



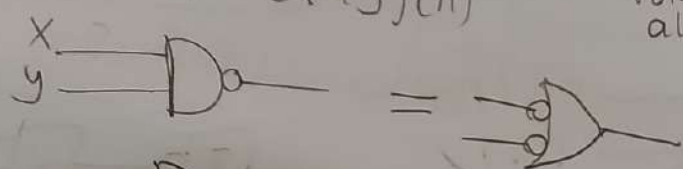
Continguate Gate Symbol

(Alternate Gate Symbol)

De Morgan

$$(\overline{X \cdot Y}) = (\overline{X} + \overline{Y})$$

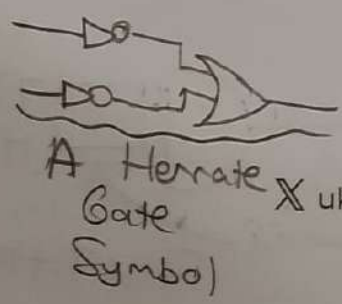
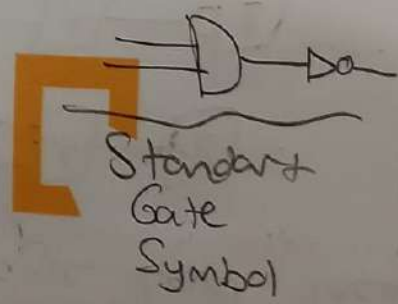
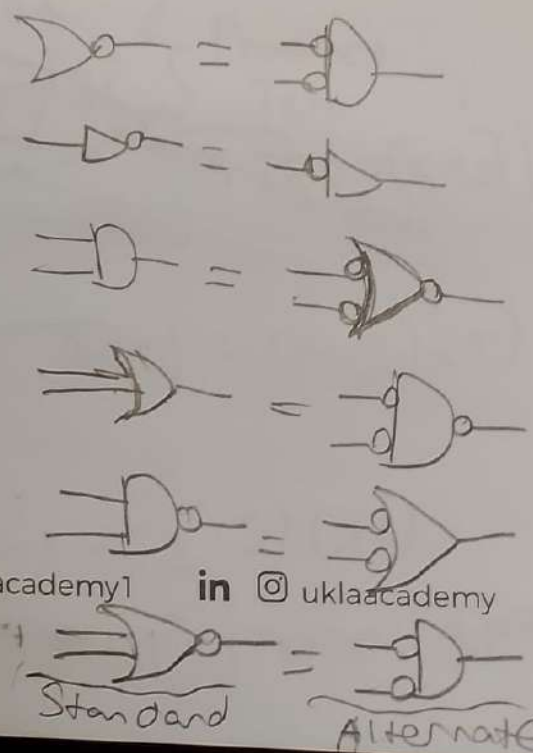
$$(X \cdot Y)(L) = (\overline{X} + \overline{Y})(H)$$



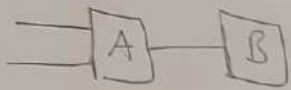
1. Operatör değiştiriliyor (dualite)
2. Giriş ve çıkışların temleyeleri alınıyor.

De Morgan

$$\overline{X+Y} = \overline{X} \cdot \overline{Y}$$



Logic Level Incompatibility Incompatibility Indicator flag



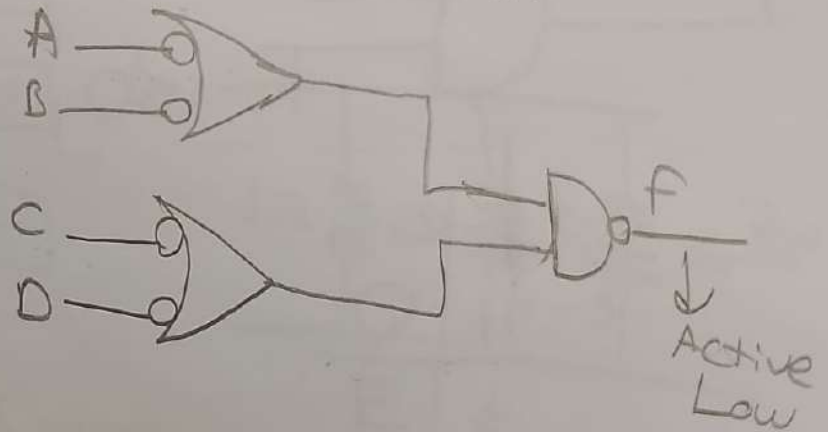
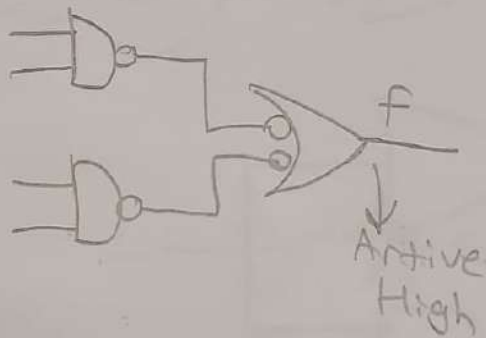
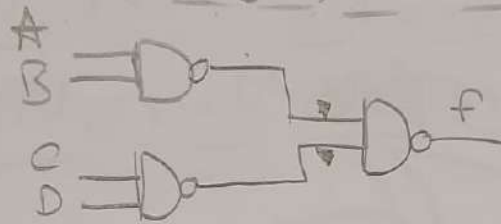
$$Z(H) = (\bar{x} \cdot y)(H)$$



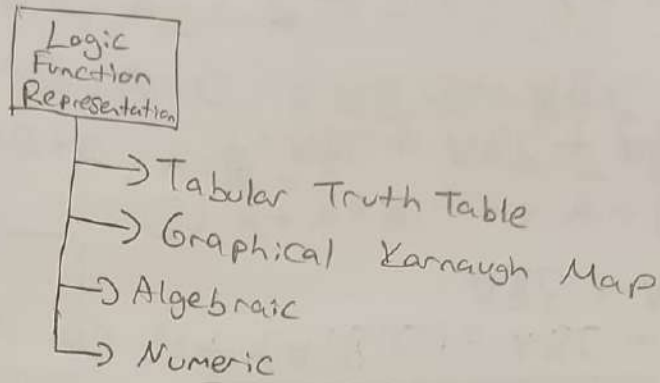
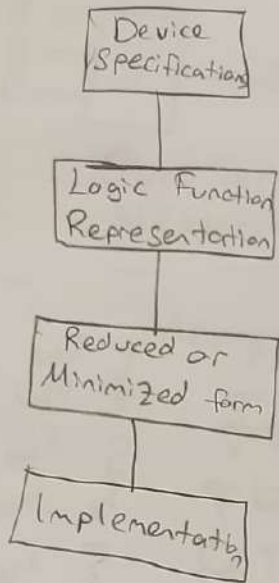
$$Z(L) = (x + \bar{y})(L)$$

Dec	ABCD	F
0	0000	0
1	0001	0
2	0010	0
3	0011	1
4	0100	0
5	0101	0
6	0110	0
7	0111	1
8	1000	0
9	1001	0
10	1010	0
11	1011	1
12	1100	1
13	1101	1
14	1110	1
15	1111	1

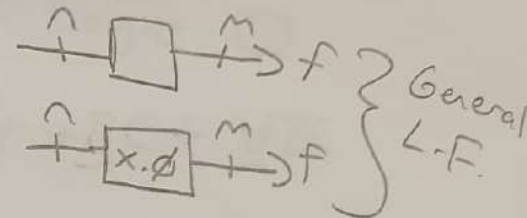
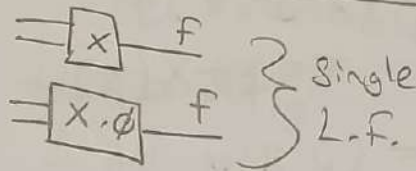
$$Z(L) = (x + \bar{y})(L)$$



LOGIC FUNCTION



Logic Function Types of Logic Function



Algebraic Representation

- SOP Sum of Products
- POS Product of Sum

SOP Representation

Product Term, minterm



• A, B, C Literal

• AB, AC, BC, $\bar{A}B$, $A\bar{C}$, $\bar{B}\bar{C}$

• ABC, $\bar{A}BC$, $A\bar{B}C$, ..., $\bar{A}\bar{B}\bar{C}$

Product Term

Minterm

minterm = Canonical Product Term =
= Fundamental

minterm Code

Uncomplemented Variable x logic 1

Complemented Variable $\bar{x}$ logic 0

$$m_i = \bar{A}BC = m_3$$

(011) $\nearrow$

* Canonical SOP form $\rightarrow$ Eger Cakpin terimlerin tomo minterm ise

Q: $f(A, B, C) = \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}BC + \bar{A}BC + ABC$

Canonical SOP form (010) + (110) + (011) + (111)

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 2 6 3 7
 $m_2 + m_6 + m_3 + m_7$

Numerical $\leftarrow$

Representation = $\sum m(2, 3, 6, 7)$

De	A	B	C	f
m_0 0	0	0	0	0
m_1 1	0	0	1	0
m_2 2	0	1	0	1
m_3 3	0	1	1	1
m_4 4	1	0	0	0
m_5 5	1	0	1	0
m_6 6	1	1	0	1
m_7 7	1	1	1	1

For Generating the Canonical SOP form

$$f(A, B, C) = A + \bar{B}C + \bar{A}BC$$

$$1) X + \bar{X} = 1$$

$$f(A, B, C) = AC(B + \bar{B}) \cdot (C + \bar{C}) + (A + \bar{A})(\bar{B}C) + \bar{A}BC$$

2) Distributive Laws

$$f(A, B, C) = ABC + A\bar{B}\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}C + \bar{A}BC$$

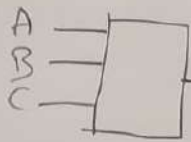
$$3) X + X = X$$

$$f = ABC + A\bar{B}\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC$$

$$f = \sum m(7, 6, 5, 4, 1, 3)$$

$$f = \sum m(2, 3, 4, 5, 6, 7)$$

POS (Product of Sum) Representation



Sum Term
Mexterm

• $A, B, C, \bar{A}, \dots, \bar{C}$ Literals

• $A+B, A+C, B+C, \bar{A}+B, \dots, \bar{B}+\bar{C}$

• $A+B+C, A+\bar{B}+\bar{C}, \dots, \bar{A}+\bar{B}+\bar{C}$

} Sum Term
Mexterm

Mexterm $M_i \Rightarrow$ Maxterm Code

Complemented variable $\bar{x}$ logic 1

Uncomplemented variable x logic 0

$$\bar{A} + \bar{B} + C + \bar{D} = M_{13}$$

$$(1 \ 1 \ 0 \ 1)_2$$

Q: $M_5 = A\bar{B}C$

$$\bar{M}_5 = \overline{A\bar{B}C} = \bar{A} + B + \bar{C}$$

M_5

$$\bar{m}_i = M_i$$

$$\bar{M}_i = m_i$$

Canonical POS form

$$f(A, B, C) = (A+B+\bar{C}) \cdot (\bar{A}+B+\bar{C}) \cdot (\bar{A}+B+C) \cdot (A+B+C)$$

$$(0 \ 0 \ 1)_2 \quad (1 \ 0 \ 1) \quad (1 \ 0 \ 0) \quad (0 \ 0 \ 0)$$

$$M_1 \cdot M_5 \cdot M_4 \cdot M_0$$

Number

$$= \Pi_m(0, 1, 4, 5)$$

Dec	ABC	F
0	0 0 0	0
1	0 0 1	0
2	0 1 0	1
3	0 1 1	1
4	1 0 0	0
5	1 0 1	0
6	1 1 0	1
7	1 1 1	1